



Index Number:

NATIONAL CERTIFICATE OF EDUCATION

2023

PHYSICS (N530)

TIME: 45 MINUTES

Candidates answer on the Question Paper.

Additional Materials: Mathematical set, Calculator

READ THESE INSTRUCTIONS FIRST

1. Write your index number in the space provided above.
2. Write in dark blue or black ink. Do not use correction fluid.
3. You may use a soft pencil for any diagram, graph or rough working.
4. Diagrams are not drawn to scale unless otherwise specified.
5. Any rough working should be done in this booklet.
6. Answer **ALL** questions.
7. This document consists of **6** questions printed on **14** pages, numbered **2** to **15**.
8. The total marks for each question is shown in brackets () at the beginning of each question.
9. The number of marks for each part question is shown in brackets [].
10. The total number of marks for this paper is **50**.

For Examiners' use								
Question No.	1	2	3	4	5	6	Total	Signature
Examiner								
Team Leader								
CE/ACE								

Question 1 (10 marks)

(a) Which instrument is used to measure **mass**?



A A measuring tape



B An electronic balance



C A stopwatch

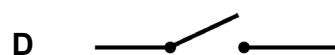
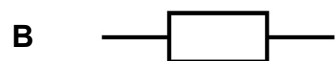
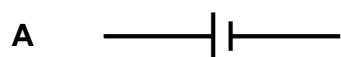


D A ruler

(b) Which one of the following is a **non-renewable** source of energy?

- A** Coal
- B** Falling water
- C** Solar energy
- D** Wind

(c) Which electrical symbol represents an **open switch**?



(d) What is the S.I. unit of **length**?

A kelvin (K)

B kilogram (kg)

C metre (m)

D second (s)

(e) Fig. 1.1 shows a **clinical thermometer**.

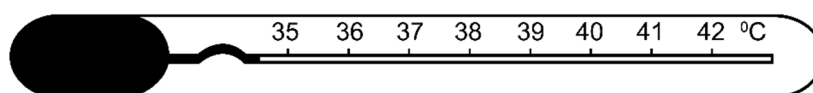


Fig. 1.1

Which temperature can be measured using this thermometer?

A 17 °C

B 27 °C

C 37 °C

D 47 °C

- (f) Fig. 1.2 shows an object **P** placed in front of a plane mirror.
Circle the letter which shows the correct position of the image formed.

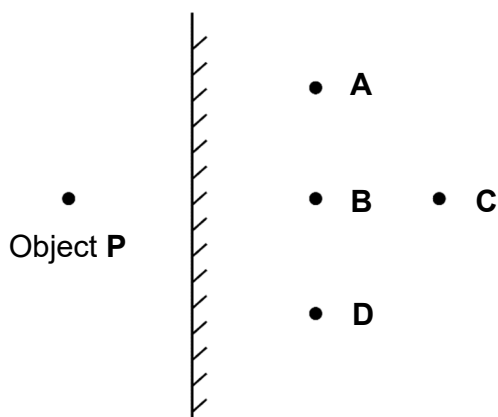
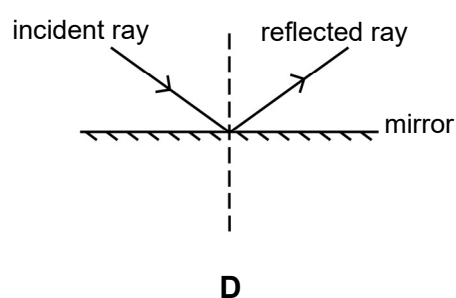
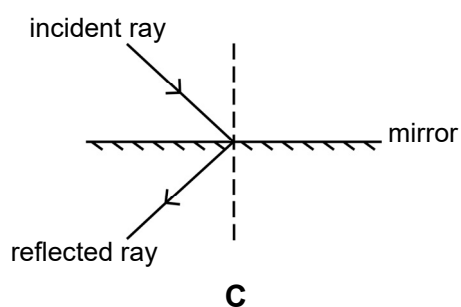
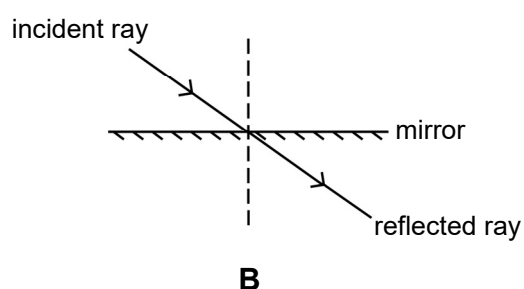
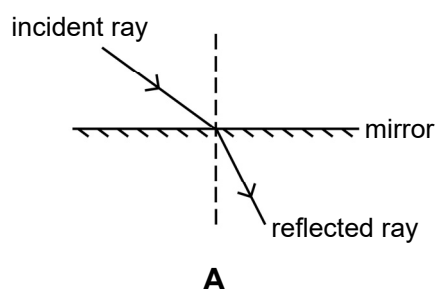


Fig. 1.2

- (g) Which one of the following is a **vector quantity**?
- A** Acceleration
 - B** Distance
 - C** Mass
 - D** Speed
- (h) Which quantity is defined as the **distance travelled per unit time**?
- A** Acceleration
 - B** Displacement
 - C** Speed
 - D** Velocity

- (i) Which one of the following ray diagrams shows the **correct** reflection of light in a plane mirror?



- (j) Fig. 1.3 shows a regular pencil.

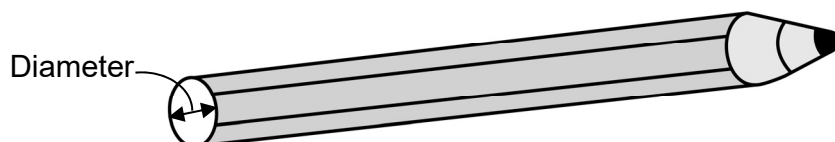


Fig. 1.3

Which one of the following is likely to be the **diameter of the pencil**?

- A** 6 km
- B** 6 m
- C** 6 cm
- D** 6 mm

E
TL
CE

Question 2 (7 marks)

Fig. 2.1 below shows a vernier caliper.

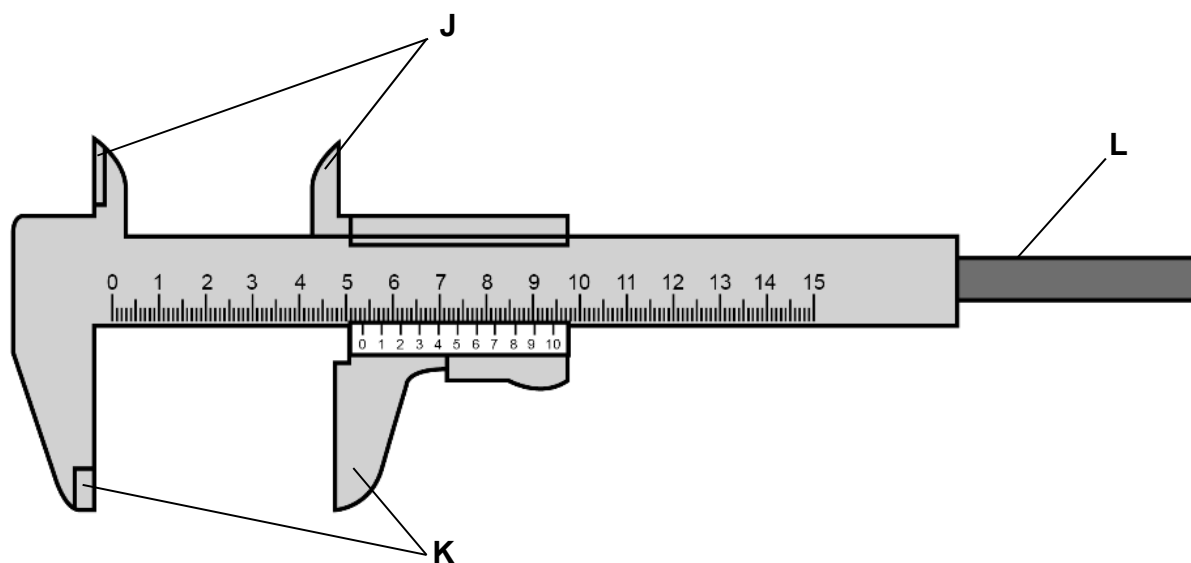


Fig. 2.1

(a) On Fig. 2.1, label the

- (i) main scale with the letter **M**.
- (ii) vernier scale with the letter **V**.

[2]

(b) Tick (✓) the correct box.

(i) What is part **J**?

☐

The internal jaws

☐

The external jaws

☐

The tail

(ii) What is part **L**?

☐

The internal jaws

☐

The external jaws

☐

The tail

[2]

(c) Tick (✓) the correct box.

Which part of the vernier caliper should be used to measure

(i) the **diameter** of a metal rod?

☐

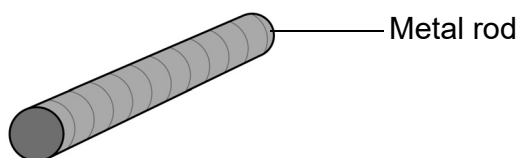
J

☐

K

☐

L



(ii) the **depth** of a beaker?

☐

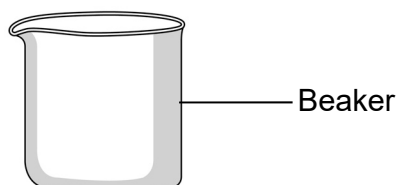
J

☐

K

☐

L



[2]

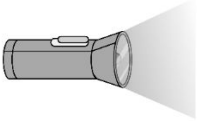
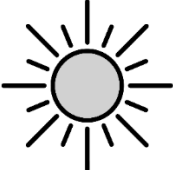
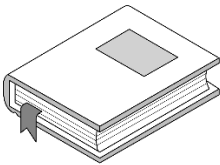
(d) Give one type of error that can occur when using a vernier caliper.

..... [1]

Question 3 (8 marks)

(a) The table below shows some objects.

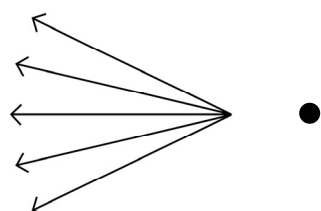
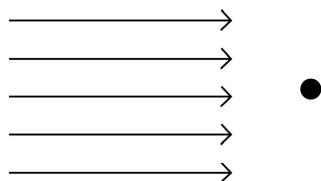
By putting a tick (✓) in the correct column, classify the objects as luminous or non-luminous.

Objects	Luminous	Non-luminous
 Lighted torch		
 Moon		
 Plane Mirror		
 Sun		
 Book		

[5]

- (b) Match each beam of light in **Column A** to its corresponding description in **Column B**.

Column A



Column B

● Divergent beam

● Convergent beam

● Parallel beam

[2]

- (c) Fig. 3.1 shows a man looking at the light from a candle through a bent flexible paper tube.

Tick (✓) the correct reason why the man cannot see the light from the candle.

☐

Light is reflected.

☐

Light travels in a straight line.

☐

Light is refracted.

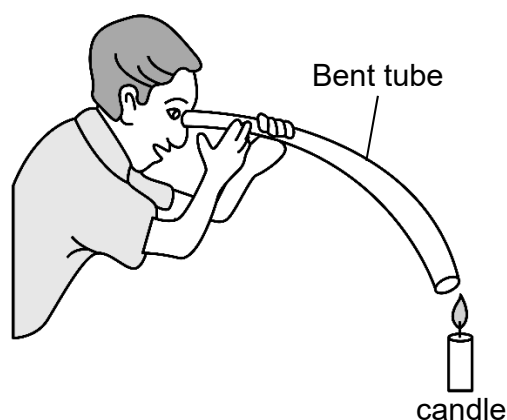


Fig 3.1

[1]

Question 4 (7 marks)

(a) Fig. 4.1 shows a circuit diagram.

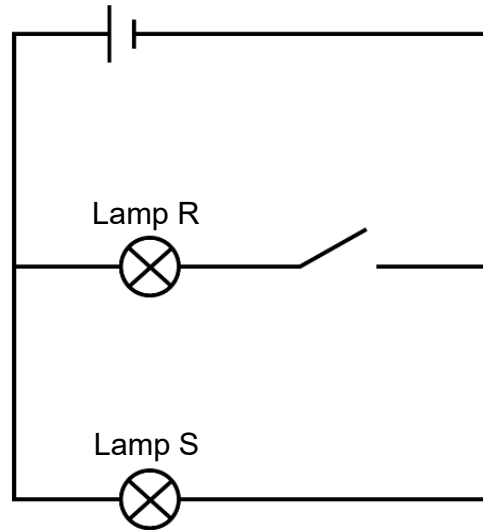


Fig. 4.1

Tick (✓) the appropriate column in the table below to indicate which lamp(s) will light up or not.

Lamp	Will light up	Will not light up
Lamp R		
Lamp S		

[2]

(b) Explain your answer in (a).

.....

..... [1]

(c) An ammeter is used to measure the current flowing in a lamp.

(i) Which electrical circuit in Fig. 4.2 shows the correct position of the ammeter?

Tick (✓) the correct box.

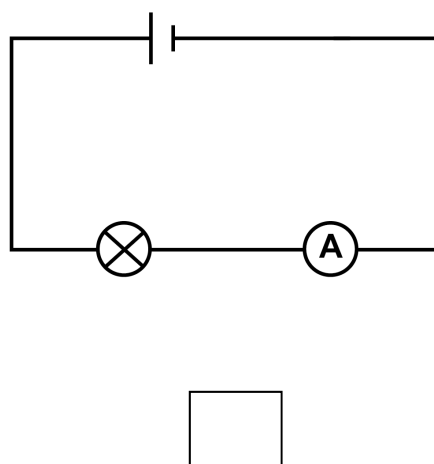
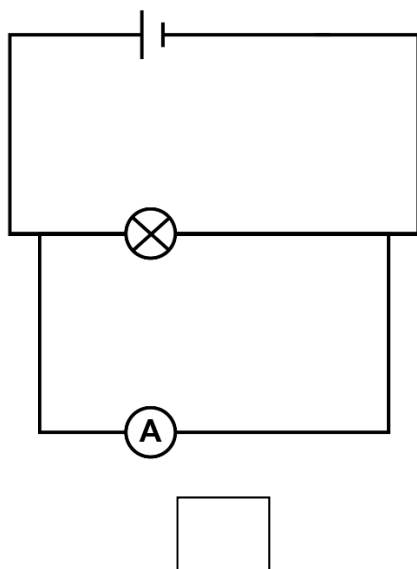


Fig. 4.2

[1]

(ii) Give a reason for your answer in (c)(i).

.....

..... [1]

(d) A current of 0.5 A flows in a circuit for 8 seconds.

Calculate the amount of charge that flows in the circuit.

Show all your workings.

Charge flowing = C [2]

Question 5 (12 marks)

- (a) Fig. 5.1 shows 3 different positions (**K**, **L**, **M**) of an apple falling from a tree.

In which position does the apple have **minimum**

- (i) potential energy?

..... [1]

- (ii) kinetic energy?

..... [1]

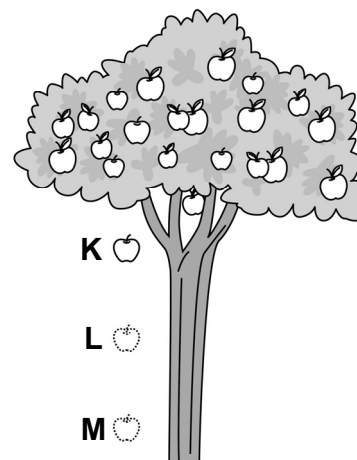


Fig. 5.1

- (b) Fig. 5.2 below shows the different positions of a pendulum bob while it is swinging to and fro between **X** and **Z**.

- (i) On Fig. 5.2, draw **two** arrows to show the direction of motion of the bob during one complete oscillation from **X**. [1]

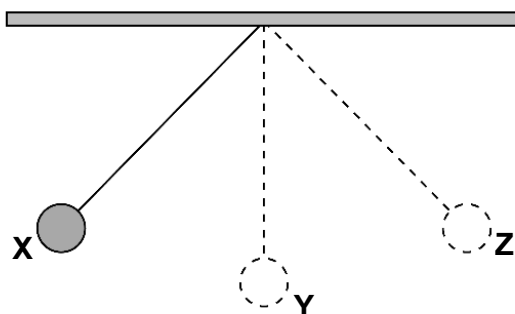


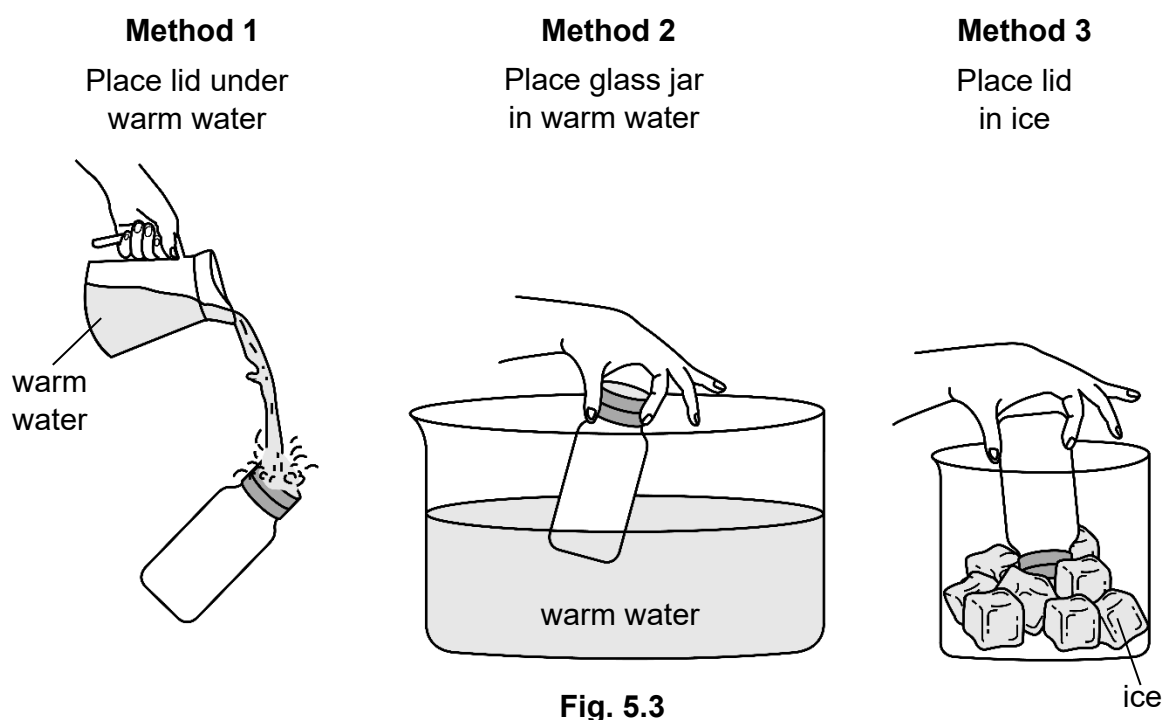
Fig. 5.2

- (ii) The pendulum bob makes 40 complete oscillations in 30 seconds.
Calculate the period of the pendulum.

Period = s [2]

- (c) A glass jar has a metal lid which is too tight to open.

A student tries to open the jar using three different methods, as shown in Fig. 5.3.



- (i) Which method will allow the student to open the lid?
Tick (✓) the correct box.

☐ Method 1

☐ Method 2

☐ Method 3

[1]

- (ii) Give a reason for your answer in (c)(i).

.....

..... [1]

- (d) The volume of two stones of different sizes is measured using the set-up shown in Fig. 5.4.

- (i) Name the method used in Fig. 5.4.

.....

[1]

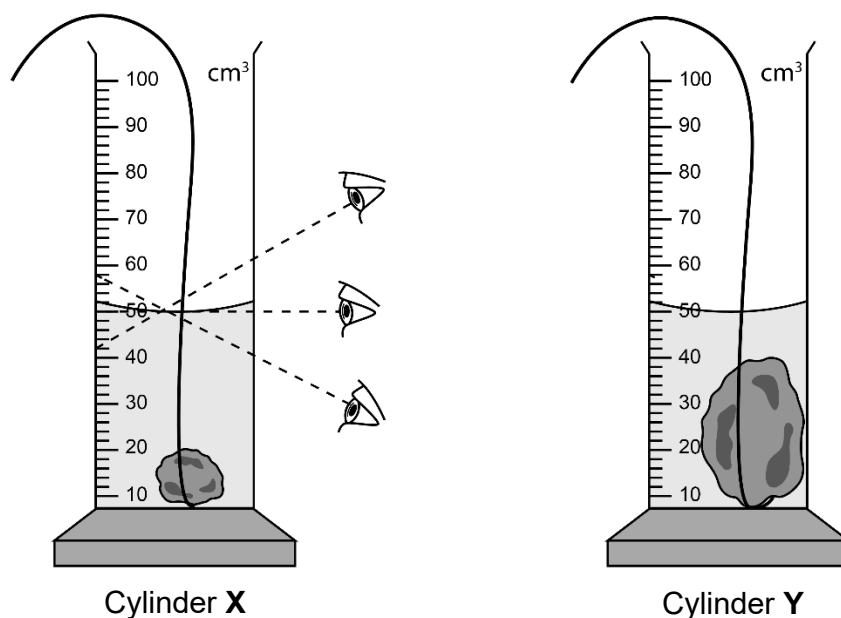


Fig 5.4

- (ii) Cylinder **X** and Cylinder **Y** **initially** contained different volumes of water.

After the stones are completely immersed, the level of water in both cylinders is the same.

Which cylinder contained the greater volume of water **at first**?

Tick (✓) the correct box.

☐

Cylinder **X**

☐

Cylinder **Y**

[1]

- (iii) Give a reason for your answer to (d)(ii).

.....

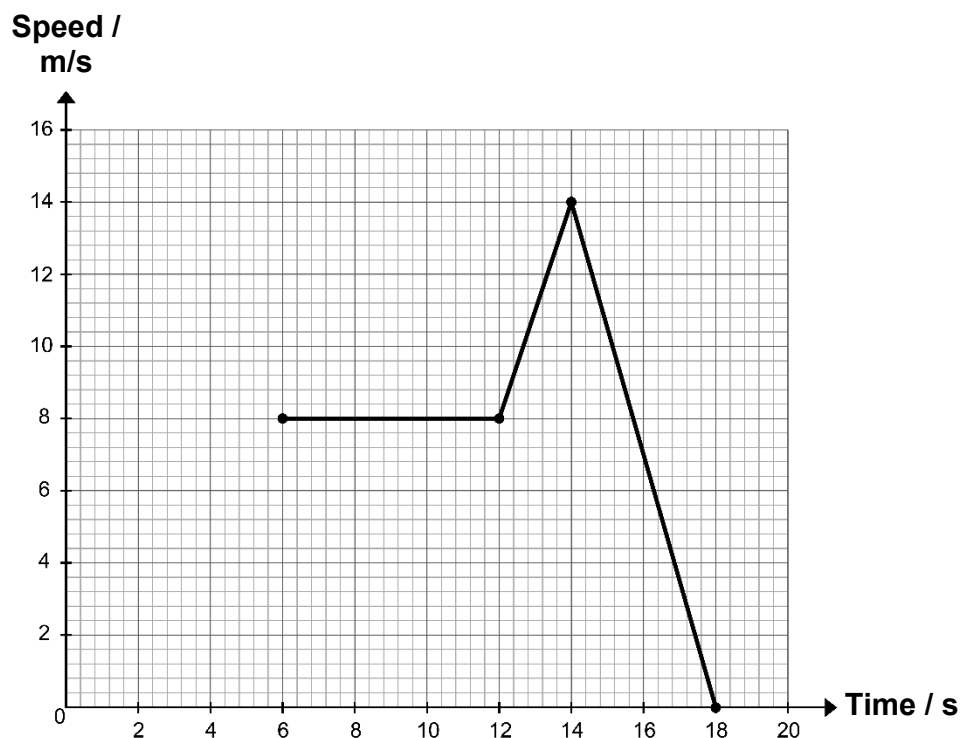
..... [2]

- (iv) On Fig. 5.4, put a tick (✓) beside the correct eye position when reading the volume in cylinder **X**.

[1]

Question 6 (6 marks)

The speed-time graph below shows part of the motion of a car moving along a straight line.



- (a) The car accelerates uniformly from **rest** to a speed of 8 m/s in the **first 6 s**.
Complete the speed-time graph to show this acceleration. [1]
- (b) For which time interval does the car move with **constant speed**? Tick (✓) the correct box. [1]
- ☐ 6 s – 12 s ☐ 12 s – 14 s ☐ 14 s – 18 s
- (c) Shade the area under the graph that represents the **distance moved during the deceleration of the car**. [1]
- (d) Calculate the **maximum** acceleration of the car.
Show all your workings.

Acceleration = m/s² [3]

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